THE TRANSFORMATIONS F STAR 1_X AND 1_X STAR F ARE HOMOTOPIC MAPS OF SIMPLICIAL OBJECTS COMPATIBLE WITH F

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: conditions (3), (4) and (5) (source lines 6802-6805) are 'checked in exactly the same manner using the strategy of the proof of Lemma godement-two-maps' - that proof (delivered as card L46158) verifies the same five conditions for the analogous family a_{i-1} star b_{n-i} with every manipulation displayed, and the substitution a_{i-1} = 1, b_{n-i} = f together with the centrality identity proven in-span at lines 6761-6766 makes each remaining check a mechanical instance of it; the footnote (lines 6807-6816) provides the independent invertible-case derivation; the author's 'We omit the details' covers only these three template instances.

Editorial note. The label lemma-godement-before-after names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Godement datum satisfying the two identities. Same chapter, \label{example-godement} and \label{lemma-godement} (card L46159 of this batch). Lines 6750-6753: the simplicial endofunctor X = (X_n, d^n_j, s^n_j) with augmentation epsilon.
- The homotopy relations. Same chapter, \label{lemma-relations-homotopy} and Remark \label{remark-homotopy-better}. Lines 6767-6770: the five conditions on the family h_{n,i}.
- The interleaving calculus for d^n_j and s^n_j. Same chapter, \label{lemma-godement-two-maps} (card L46158 of this batch), its proof. Lines 6772-6774: the rewriting formulas for d^n_j and s^n_j are used for the remaining conditions.
- Centrality of endomorphisms of the identity under the star product. Same chapter, \label{lemma-godement-two-maps}-style identity, proven in-span. Lines 6761-6766: a circ (f star 1_F) = f star a = (f star 1_G) circ a for any transformation a.

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0G5M: sources/stacks-simplicial/pinned-original-source.tex, source lines 6410–6410. Prerequisite pointer invoked at proof line(s) 6752; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G5N: sources/stacks-simplicial/pinned-original-source.tex, source lines 6443–6443. Prerequisite pointer invoked at proof line(s) 6753; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G5R: sources/stacks-simplicial/pinned-original-source.tex, source lines 6612–6612. Prerequisite pointer invoked at proof line(s) 6785; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 019L: sources/stacks-simplicial/pinned-original-source.tex, source lines 4704–4704. Prerequisite pointer invoked at proof line(s) 6775; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 019M: sources/stacks-simplicial/pinned-original-source.tex, source lines 4737–4737. Prerequisite pointer invoked at proof line(s) 6776; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$, Y, d, s be as in Example 0G5M satisfying the equations of Lemma 0G5N. Let $f : id_{\mathcal{C}} \to id_{\mathcal{C}}$ be an endomorphism of the identity functor. Then $f \star 1_X, 1_X \star f : X \to X$ are maps of simplicial objects compatible with f via the augmentation $\epsilon : X \to id_{\mathcal{C}}$. Moreover, $f \star 1_X$ and $1_X \star f$ are homotopic.*

Proof. The map $f \star 1_X$ is the map with components

$$X_n = \mathrm{id}_{\mathcal{C}} \circ X_n \xrightarrow{f \star 1_{X_n}} \mathrm{id}_{\mathcal{C}} \circ X_n = X_n$$

For a transformation $a : F \to G$ of endofunctors of $\mathcal{C}$ we have $a \circ (f \star 1_F) = f \star a = (f \star 1_G) \circ a$. Thus $f \star 1_X$ is indeed a morphism of simplicial objects. Similarly for $1_X \star f$.

To show the morphisms are homotopic we construct morphisms $h_{n,i} : X_n \to X_n$ for $n \geq 0$ and $0 \leq i \leq n+1$ satisfying the relations described in Lemma 019L. See also Remark 019M. It turns out we can take

$$h_{n,i} = 1_i \star f \star 1_{n+1-i}$$

where 1_i is the identity transformation on $Y \circ \ldots \circ Y$ as in the proof of Lemma 0G5N. We have $h_{n,0} = f \star 1_{X_n}$ and $h_{n,n+1} = 1_{X_n} \star f$ which checks the first condition. In checking the other conditions we use the comments made in the proof of Lemma 0G5R about the maps d^n_j and s^n_j.

We check condition (2) of Lemma 019L. Let $i > j$. We have to show

$$d^n_j \circ (1_i \star f \star 1_{n+1-i}) = (1_{i-1} \star f \star 1_{n+1-i}) \circ d^n_j$$

Since $i - 1 \geq j$ we can use one of the possible descriptions of d^n_j to rewrite the left hand side as

$$(d^{i-1}_j \star 1_{n-i+1}) \circ (1_i \star f \star 1_{n+1-i}) = d^{i-1}_j \star f \star 1_{n+1-i}$$

Similarly the right hand side becomes

$$(1_{i-1} \star f \star 1_{n+1-i}) \circ (d^{i-1}_j \star 1_{n-i+1}) = d^{i-1}_j \star f \star 1_{n+1-i}$$

Thus we obtain the same result and (2) is checked.

The conditions (3), (4), and (5) of Lemma 019L are checked in exactly the same manner using the strategy of the proof of Lemma 0G5R. We omit the details[1]. □

[1]When f is invertible it suffices to prove that $(a_n) = 1_X$ and $(b_n) = f^{-1} \star 1_X \star f$ are homotopic. But this follows from Lemma 0G5R because in this case $a = b = 1_{\mathrm{id}_\mathcal{C}}$.